

Scientific Computation

Prof. Matthew Giamou
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Lecture 15 - Iterative Methods

Lecture Outline

- Positive Definiteness Cont.

- The Cholesky Decomposition
- Iterative Methods

Positive (Semi-)Definite Matrices

A matrix $A \in \mathbb{R}^{n \times n}$ is **positive (semi-)definite** if

$$x^T A x > 0 \quad (x^T A x \geq 0)$$

for all nonzero $x \in \mathbb{R}^n$. We use the following “curly” inequality to indicate this:

$$A \succ 0 \quad (A \succeq 0),$$

and if A is symmetric in addition to positive (semi-)definite, we write

$$A \in \mathbb{S}_{++}^n \quad (A \in \mathbb{S}_+^n).$$

Eigenvalues and Positive (Semi-)Definiteness

There is another useful characterization of PD and PSD matrices:

$$x^T A x > 0 \quad \forall x \neq 0$$

→ all of A 's eigenvalue are > 0 .

otherwise let $x_i =$ eigenvector for eigenvalue $\lambda_i < 0$

$$x_i^T A x_i = x_i^T \lambda_i x_i = \lambda_i \|x_i\|_2^2 = \lambda_i \sum_{j=1}^n [x_i(j)]^2 < 0$$

which contradicts $A > 0$ (similar for $A \geq 0$)

Diagonal Dominance

A matrix is **diagonally dominant** if the absolute value of the diagonal element in each row is greater than the sum of the absolute value of the other elements in the same row:

$$\sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}| \leq |a_{ii}|, \quad i = 1 \dots n$$

↳ strictly diag. dom. if $<$ instead

If $A \in \mathbb{R}^{n \times n}$ and $a_{ii} > 0$, and A is diag. dominant, then $A > 0$

But, PD $\neq$ diag. dom.

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- Positive Definiteness
- **The Cholesky Decomposition**
- Iterative Methods

The Cholesky Factorization

The symmetric LDL^T factorization has a special case when $A \succ 0$:

$$x^T A x = x^T L D L^T x = (L^T x)^T D L^T x > 0$$

$$\text{Let } y \triangleq L^T x \Rightarrow y^T D y = \sum_{i=1}^n d_i y_i^2 > 0$$

$$\Rightarrow d_i > 0, i = 1, \dots, n$$

$$G \triangleq L \begin{bmatrix} \sqrt{d_1} & & 0 \\ & \ddots & \\ 0 & & \sqrt{d_n} \end{bmatrix} \Rightarrow G G^T = L D L^T = A$$

$\hookrightarrow$ lower triangular Cholesky factor.

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Iterative Methods

So far, we have studied **direct methods** (i.e., based on Gaussian elimination) for solving linear systems

↳ are all $O(n^3)$ runtime and we use $O(n^2)$ storage.
what if n really large.

Eg. for large sparse matrix where the number of non zero entries of A is $\ll n^2$ for very large n .

3x3 Example

Let's look at the simple case where $A \in \mathbb{R}^{3 \times 3}$:

$$\begin{aligned} Ax = b \quad \Leftrightarrow \quad x_1 &= (b_1 - a_{12} \cdot x_2 - a_{13} \cdot x_3) / a_{11} \\ x_2 &= (b_2 - a_{21} x_1 - a_{23} x_3) / a_{22} \\ x_3 &= (b_3 - a_{31} x_1 - a_{32} x_2) / a_{33} \end{aligned}$$

require knowledge of the
other entries of x .

3x3 Example Cont.

Now let's assume we're at iteration k and have access to the previous iteration's approximate solution $x^{(k-1)}$:

$$x_1^{(k)} = (b_1 - a_{12}x_2^{(k-1)} - a_{13}x_3^{(k-1)}) / a_{11}$$

$\vdots$

$$x_3^{(k)} = (b_3 - a_{31}x_1^{(k-1)} - a_{32}x_2^{(k-1)}) / a_{33}$$

$\Rightarrow$ we're updating $x^{(k)}$ using $x^{(k-1)}$

The Jacobi Method

We can extrapolate this to $A \in \mathbb{R}^{n \times n}$ in the following simple loop of **Jacobi iterations**:

Given $x^{(0)}$:

for $k=1 \dots k_{\max}$

for $i=1 \dots n$

$$x_i^{(k)} = (b_i - \sum_{j=1}^n a_{ij}x_j^{(k-1)}) / a_{ii}$$

j=i

Termination criteria, e.g. $\|Ax^{(k)} - b\| \leq \text{tolerance}$

end

We sometimes guess the previous values of x

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Matrix Interpretation

Each iteration can be succinctly understood as matrix multiplications:

$$A = L_A + D_A + U_A$$

$$= \begin{bmatrix} 0 & & & 0 \\ a_{21} & \ddots & & \\ \vdots & \ddots & \ddots & \\ a_{n1} & & & 0 \end{bmatrix} + \begin{bmatrix} a_{11} & & & 0 \\ & a_{22} & & \\ & & \ddots & \\ 0 & & & a_{nn} \end{bmatrix} + \begin{bmatrix} 0 & a_{12} & \dots & a_{1n} \\ & \ddots & \ddots & \vdots \\ & & \ddots & a_{n-1,n} \\ 0 & & & 0 \end{bmatrix}$$

$$\text{Then } x^{(k)} = D_A^{-1} (b - \underbrace{(L_A + U_A)x^{(k-1)}})$$

This is just the sum in
our Jacobi iteration

i.e. $-Ax^{(k-1)}$ with the diag. part removed

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Gauss-Seidel

What if we use the most up-to-date value for $x_i^{(k)}$?

Given $x^{(0)}$:

for $k=1 \dots K_{\max}$

for $i=1 \dots n$

$$x_i^{(k)} = (b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k)} - \sum_{j=i+1}^n a_{ij} x_j^{(k-1)}) / a_{ii}$$

end

Terminate criteria

end

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Matrix Interpretation of Gauss-Seidel Iterations

We can once again describe Gauss-Seidel iterations as a matrix product

The vector with entries $\sum_{j=1}^{i-1} a_{ij} x_j^{(k)}$ is simply $L_A x^{(k)}$,

$$\text{and } \sum_{j=i+1}^n a_{ij} x_j^{(k-1)} = U_A x^{(k-1)}$$

So we have

$$x^{(k)} = D_A^{-1} (b - L_A x^{(k)} - U_A x^{(k-1)})$$

$$(D_A^{-1} L_A + I) x^{(k)} = D_A^{-1} (b - U_A x^{(k-1)})$$

$$(L_A + D_A) x^{(k)} = b - U_A x^{(k-1)}$$

$$x^{(k)} = (L_A + D_A)^{-1} (b - U_A x^{(k-1)})$$

Splitting A Into M and N

We have just shown two “splittings” of A :

$$A = M - N$$

$$D_A x^{(k)} = \underbrace{(-L_A - U_A)}_{N_J} x^{(k-1)} + b$$

for Jacobi iterations, and

$$\underbrace{(L_A + D_A)x^{(k)}}_{M_{GS}} = - \underbrace{U_A x^{(k-1)}}_{N_{GS}} + b$$

$$A = M_{GS} - N_{GS}$$

$$= L_A + D_A + U_A$$

for Gauss-Seidel.

$$Mx^{(k)} = Nx^{(k-1)} + b$$

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We are interested in characterizing the **convergence** of iterative methods of the form

$$Mx^{(k)} = Nx^{(k-1)} + b.$$

Specifically, what are the requirements on M and N (and therefore A) such that

$$\lim_{k \rightarrow \infty} x^{(k)} = x^* \triangleq A^{-1}b?$$

Let $e^{(k)} = x^{(k)} - x^*$, and define $\theta \triangleq n^{-1}N$. Note that $Nx^* = N\bar{x}^* + b$

(because $A = M - N$). Then, $M(x^{(k)} - x^*) = Nx^{(k-1)} + b - Nx^* - b$
 $= N(x^{(k-1)} - x^*)$

$$\text{So } Me^{(k)} = Ne^{(k-1)} \Rightarrow e^{(k)} = M^{-1}Ne^{(k-1)} = Ge^{(k-1)}$$

$$\|e^{(k)}\| = \|Ge^{(k-1)}\| = \|G^k e^{(0)}\| \leq \|G^k\| \cdot \|e^{(0)}\|$$

$$\Rightarrow \|G\| < 1 \text{ then } e^{(k)} \rightarrow 0$$

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Convergence Theorems

The Jacobi and Gauss-Seidel methods have the following convergence results (which we will not prove):

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关于 Jacobi 迭代法:

✿ 一、背景：我们要求解的是线性方程组

$$Ax = b$$

也就是：

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1, \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2, \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3. \end{cases}$$

二、Jacobi 迭代的想法

把每个方程中的「对角项」单独解出来，
比如第 i 个方程：

$$a_{ii}x_i = b_i - \sum_{j \neq i} a_{ij}x_j$$

于是可以写成：

$$x_i = \frac{1}{a_{ii}} \left(b_i - \sum_{j \neq i} a_{ij}x_j \right)$$

三、Jacobi 迭代形式

我们不知道精确解，所以用上一次的近似值 $x^{(k-1)}$ 去更新：

$$x_i^{(k)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j \neq i} a_{ij}x_j^{(k-1)} \right)$$

也就是说：

第 k 次迭代的结果，完全用第 $(k-1)$ 次迭代的值来计算。

四、图中写的公式对应的是 3×3 的情况

图上写的就是这三个公式的展开：

$$\begin{aligned} x_1^{(k)} &= \frac{1}{a_{11}} (b_1 - a_{12}x_2^{(k-1)} - a_{13}x_3^{(k-1)}) \\ x_2^{(k)} &= \frac{1}{a_{22}} (b_2 - a_{21}x_1^{(k-1)} - a_{23}x_3^{(k-1)}) \\ x_3^{(k)} &= \frac{1}{a_{33}} (b_3 - a_{31}x_1^{(k-1)} - a_{32}x_2^{(k-1)}) \end{aligned}$$

最后一句 “we’re updating $x^{(k)}$ using $x^{(k-1)}$ ”
意思就是：

我们用上一轮的近似解来更新当前一轮的近似解。↓

💡 **五、那 $x^{(k-1)}$ 怎么确定？**

这个问题非常关键！

在迭代法中，**必须从一个初始猜测值开始**，
记为：

$$x^{(0)} = \begin{bmatrix} x_1^{(0)} \\ x_2^{(0)} \\ x_3^{(0)} \end{bmatrix}$$

这个 $x^{(0)}$ 是我们“第 0 次迭代”的近似解。

常见的几种选法：

初始向量	含义	备注
$x^{(0)} = 0$	全零向量	最常用、简单
$x^{(0)} = b$	用右端项	有时更快
上一次求解结果	当要解相似系统时	在连续问题中常用

🔄 **六、迭代过程示意**

- 1 选择初始值 $x^{(0)}$
- 2 用 Jacobi 公式计算 $x^{(1)}$
- 3 再用 $x^{(1)}$ 计算 $x^{(2)}$
- ...

直到：

$$\|x^{(k)} - x^{(k-1)}\| < \text{容差}$$

或残差：

$$\|Ax^{(k)} - b\| < \text{容差}$$

Gauss-Seidal:

🧠 二、把 A 拆成三部分

在迭代法里，我们把 A 拆成三块：

$$A = L_A + D_A + U_A$$

其中：

- L_A ：下三角部分（不含对角线）

$$L_A = \begin{bmatrix} 0 & 0 & 0 \\ a_{21} & 0 & 0 \\ a_{31} & a_{32} & 0 \end{bmatrix}$$

- D_A ：对角矩阵 (Diagonal)

$$D_A = \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix}$$

- U_A ：上三角部分（不含对角线）

$$U_A = \begin{bmatrix} 0 & a_{12} & a_{13} \\ 0 & 0 & a_{23} \\ 0 & 0 & 0 \end{bmatrix}$$

🧩 三、把 $Ax = b$ 改写成迭代形式

方程：

$$(L_A + D_A + U_A)x = b$$

我们希望把未知项 x 拆出来一部分放到左边（容易反解），另一部分放右边。

Jacobi 法的核心想法是：

只保留 D_A 在左边（因为它容易求逆），

把其他项 $L_A + U_A$ 放到右边。

于是：

$$D_A x = b - (L_A + U_A)x$$

🔄 四、用上一次的近似解更新

因为右边的 x 我们还不知道，所以用上一次迭代值 $x^{(k-1)}$ 来近似：

$$D_A x^{(k)} = b - (L_A + U_A)x^{(k-1)}$$

两边同乘 D_A^{-1} :

$$x^{(k)} = D_A^{-1}(b - (L_A + U_A)x^{(k-1)})$$

■ 五、解释图中每一部分

图上写的:

$$x^{(k)} = D_A^{-1}(b - (L_A + U_A)x^{(k-1)})$$

并且注释:

“This is just the sum in our Jacobi iteration, i.e. $-Ax^{(k-1)}$ with the diag part removed.”

意思是:

- $(L_A + U_A)x^{(k-1)}$ 就是上次的 $Ax^{(k-1)}$, 但把对角部分去掉。
- 因为 Jacobi 法中, 每个新 x_i 只用旧的非对角元素计算。

📖 六、对应的向量公式 (你之前见过的那种)

这个矩阵式其实就是:

$$x_i^{(k)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j \neq i} a_{ij} x_j^{(k-1)} \right)$$

🌟 七、总结一句话

这张图是在把 Jacobi 迭代法写成「矩阵形式」:

$$x^{(k)} = D_A^{-1}(b - (L_A + U_A)x^{(k-1)})$$

其中:

- “ D_A : 对角部分 (容易求逆)”
- “ L_A, U_A : 下/上三角部分, 用旧解计算”

这个公式正是 Jacobi 迭代的**矩阵版本**。